

Problem 7.17

Show that the discount function of a simple interest rate i is larger than the discount function for a compound interest rate i for $t \geq 1$.

Solution

The discount function under simple interest is

$$v_s(t) = \frac{1}{1 + it},$$

while under compound interest it is

$$v_c(t) = \frac{1}{(1 + i)^t}.$$

We want to show that

$$\frac{1}{1 + it} \geq \frac{1}{(1 + i)^t} \quad \text{for } t \geq 1.$$

Since $i > 0$, both denominators are positive, so this is equivalent to showing

$$1 + it \leq (1 + i)^t.$$

But this follows from Bernoulli's inequality, which states that for $i > -1$ and $t \geq 1$,

$$(1 + i)^t \geq 1 + it.$$

Therefore,

$$\frac{1}{(1 + i)^t} \leq \frac{1}{1 + it}.$$

Hence, the discount function for simple interest is larger than the discount function for compound interest for $t \geq 1$.

$$\boxed{\frac{1}{1 + it} \geq \frac{1}{(1 + i)^t}}.$$